Linux & shell · Quick concepts

Interview revision sheet. Read it the evening before, then test yourself in Interview practice.

Navigation and files

- pwd, ls -la, cd, cp, mv, rm, mkdir, find /path -name file.
- chmod 755 (rwx r-x r-x), chown user:group.
- less for big files; head and tail -n; tail -f to follow.

Searching and text

- grep -i (ignore case), -v (invert), -c (count), -r (recursive); zgrep for .gz files.
- awk '{print $1}' prints the first column; sed -i 's/old/new/g' edits in place.
- sort, uniq -c, sort -nr, head: top N frequent values.
- wc -l counts lines.

Processes and system

- ps -ef, top or htop, kill PID (graceful), kill -9 (force, last resort).
- uptime and load average: compare with the number of CPU cores.
- free -h (memory), df -h (disk), df -i (inodes), du -sh * (folder sizes).
- ss -tlnp or netstat -tlnp (listening ports); lsof -i :8080.

Services and logs

- systemctl status, start, stop, restart; journalctl -u service.
- logrotate rotates and compresses logs; missing rotation fills disks.
- Deleted file still using space: a process holds it open (lsof).

Networking

- ping (reachability), curl -v (HTTP), nc -zv host port (TCP port), nslookup or dig (DNS), traceroute.
- openssl s_client to check certificates and expiry.

Shell scripting

- #!/bin/bash; set -euo pipefail for safer scripts.
- > overwrites, >> appends, 2>&1 sends errors to the same place.
- $? holds the last exit code (0 = success). Quote variables: "$FILE".
- cron: minute hour day month weekday; crontab -l and -e; use full paths.
- nohup keeps a job running after logout.

Troubleshooting

- High load plus high I/O wait: disk or network bottleneck.
- OOM killer messages in dmesg or journalctl.
- ulimit -n for open file limits; strace -p PID to see system calls.